Technology & Architecture Overview

Fibre Assurance & Network Operations Platform

The platform is built on standard open-source infrastructure—Python, FastAPI, PostgreSQL/PostGIS, Kafka, and React—designed to run either in the cloud or inside an air-gapped carrier data centre. All fault pinpointing and routing run on deterministic math; language models are optional and used only for final text summaries.

System Flow

1. INGEST

Network & IT Feeds

Polls GIS, DWDM, RAN & BSS APIs; listens to live Syslog, SNMP traps & binary OTDR files.

2. STORE

Event & Spatial Data

Buffers alarm bursts in Redpanda/Kafka; stores 6-layer topology & street geometry in PostGIS.

3. COMPUTE

Physics & Graph Engine

Clusters related alarms, subtracts manhole slack on GIS maps & calculates diverse backup paths.

4. PRESENT

Web UI & Gated Actions

Shows exact street GPS, OTDR curves & SLA risk; executes reroutes/tickets after NOC approval.

Core Technology Stack

LAYER	TECHNOLOGIES	WHAT IT HANDLES
Web Frontend	`React 18`, `TypeScript 5`, `Tailwind CSS`, `Vite`	Operations console, incident workflow, OTDR trace charts, dependency visuals, and light/dark modes.
GIS Mapping	`MapLibre GL JS` (WebGL)	Renders street ducts, fibre cables, chambers, 5G sites, and raw-vs-corrected GPS dig pins at 60 fps.
Backend Engine	`Python 3.11+`, `FastAPI`, `Pydantic v2`, `HTTPX`	Async REST endpoints, typed API contracts, pluggable fault-classifier registry, and connector sync.
Spatial & Vector DB	`PostgreSQL 16`, `PostGIS 3.4`, `pgvector`	Normalized 6-layer network inventory, `WGS84` trench interpolation, and runbook vector search.
Event Streaming	`Redpanda` (Kafka API), `asyncio` UDP	High-throughput alarm ingestion (`:19092`) plus native `RFC 5424` Syslog (`:15514`) & SNMPv2c (`:15162`).
Graph & Fibre Codec	`NetworkX`, custom `SR-4731` (`.sor`) parser	Connected-subgraph multi-fault isolation, Dijkstra optical rerouting, and binary OTDR + CRC-16 parsing.
Summary AI (Optional)	`Anthropic Claude` / `OpenAI GPT`	Drafts human-readable incident briefs and RFOs from verified data; includes full offline fallback.

Supported Telecom & IT Interfaces

NETWORK DOMAIN	REPRESENTATIVE SYSTEMS	PROTOCOLS & FORMATS
Optical / DWDM	Ciena MCP, Nokia NSP	`RESTCONF`, `OpenConfig gNMI`, `TMF642`
Fibre Testing (OTDR)	VIAVI ONMSi, EXFO Nova Fiber	Binary `Telcordia SR-4731 Rev 2 (.sor)`
Outside-Plant GIS	Lepton SmartInventory, ESRI, QGIS	`OGC WFS 2.0 GeoJSON` (ducts, chambers, slack)
5G RAN & Access	Ericsson ENM, GPON OLTs / GenieACS	`3GPP TS 32.435 XML`, `ITU-T G.984` Dying-Gasp
IP Core & Subsea	Cisco, Juniper, Subsea CLS PFE	`UDP Syslog`, `SNMPv2c`, PFE & C-OTDR telemetry
ITSM, DCIM & BSS	ServiceNow, Jira SM, NetBox, Comviva	`REST v3 / Table API`, `TMF629` & `TMF638` SLAs

Engineering Architecture Guardrails

1. **Deterministic Core** — Coordinates, manhole slack subtraction, and backup wavelengths are calculated strictly by PostGIS, binary OTDR math, and NetworkX—never estimated by generative AI.
2. **Topology-Agnostic Code** — Application logic contains zero hardcoded cable, duct, or router IDs (enforced on every commit in GitHub Actions CI). Any operator's GIS dataset loads via configuration.
3. **Pluggable Classifiers** — Each fault signature (fibre cut, premises power loss, tower battery drain, rain fade, optical degradation, subsea shunt) is an isolated, unit-tested Python plugin.
4. **Built-In Lab Emulator** — Includes a standalone Multi-OEM Emulator (`:8001`) that generates real binary `.sor` traces, WFS GeoJSON, 3GPP XML, and 10k-alarm storms for safe testing and demos.
5. **Human-in-the-Loop Safety** — Southbound ingestion is strictly read-only. Optical wavelength switches and field work orders are staged as drafts and execute only after explicit NOC approval.

Local & Production Runtime: Packaged with `Docker Compose` (PostgreSQL 16 + PostGIS + pgvector on `:5433`, Redpanda Kafka on `:19092`), FastAPI backend (`:8002`), Multi-OEM lab emulator (`:8001`), and React web UI (`:5173`). Deploys unchanged to standard Kubernetes (`EKS / AKS / OpenShift`) or on-prem Linux VMs.